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Heart Disease Prediction Using ML and Ensemble Models: A Comprehensive Analysis

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***Abstract*— The leading cause of death worldwide is still heart disease, highlighting the urgent need for improved diagnostics. This research included a range of ML techniques. Employing SVM, RF, DT, KNN, and LGBM to predict heart illness. The assessment metrics consist of F1 score, recall, accuracy, ROC, and precision. To ensure accurate analysis, data pretreatment techniques including reduction, integration, transformation, and purification were applied. Hyperparameter adjustment improved the model's performance even further. Among the models analyzed, Naïve Bayes performed the worst, with an F1 score of 98%, accuracy of 69.63%, precision of 93%, and recall of 98%. Three distinct datasets were employed in the study, and their strengths were combined to enhance model discrimination, reduce overfitting, and raise overall accuracy. We combined DT, LGNM, and LGBM to create a number of models. From RF we are able to achieve an accuracy of up to 98.89%, Precision 99.00%, Recall 98.00% and F1-Score 99.00%. These results show how well ensemble methods and data integration may enhance machine learning-based heart disease prediction abilities.**

Keywords—Heart disease, Machine learning, Combined dataset, Ensemble machine learning.

I. INTRODUCTION

Heart Ailment, the world's leading cause of illness and death, accounts for more than 70% of all fatalities. The 2017 study on the global burden of disease [1] states that about 43% of fatalities globally are caused by heart disease. According to the Australian Institute of Health and Welfare (AIHW), Heart Ailment would be responsible for 42% of all deaths in Australia in 2018 [2]. Heart disease can be caused by a variety of variables, including high blood pressure, obesity, high cholesterol, smoking, drinking, not exercising, diabetes, and abnormal cardiac rhythms. However, people can control these problems. Clinical studies show that factors include weight increase, a high-fat diet, low levels of HDL cholesterol, high blood pressure, and a significant quantity of LDL (bad) cholesterol history of cardiac problems [3]. Furthermore, some customers believe that some technologies, such as electrocardiograms and CT scans, are too costly and impractical to detect coronary cardiovascular disease. The aforementioned reason alone has resulted in the deaths of 17 million people [4]. The World Health Organization (WHO) estimates that there are around 23.6 million deaths annually. By 2030, cardiovascular diseases, mainly strokes and cardiac arrests, may be the cause of deaths globally [5].

Therefore, in order to save lives, it is imperative to apply data mining and ML approaches to anticipate the likelihood of developing heart illness lessening the societal financial load. ML is vital to the medical field. We can identify, determine the best course of action, and estimate a wide range of illnesses thanks to machine learning. It has recently become more popular to use data mining and ML techniques to predict the chance of contracting specific diseases. It has previously been established that data mining techniques may be used to forecast sickness. Although some research has tried to predict the probability that the disease would get worse in the future, reliable findings have not yet been found [6]. This study presents an ensemble model that combines feature extraction and data balancing across three datasets to enhance the prediction of heart disease. The following crucial This paper presents the following work:

- Null value handling, feature scaling, data balancing, and missing data handling are used for data preparation, and hyper parameter tuning is used to optimize the hyperparameters of ML models to enhance their performance.
- This work makes use of a number of ML classifiers, including DT, RF, KNN, LGBM, SVM, and others.
- The soft voting ensemble approach was ultimately used to merge the models with the highest accuracy in order to increase the model's accuracy.
- By employing grid search for hyperparameter tuning, this study improves a ML model by optimizing variables such as learning rate, batch size, and dropout rate. This method guarantees solid, trustworthy findings while increasing accuracy and decreasing overfitting.

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II. Literature Review

ML algorithms have been extensively studied for the prediction of Heart Ailment based on clinical data. These datasets provide challenges due to their intricacy and unequal distribution of classes. If these issues are not resolved, ML models may become less accurate and efficient. To address these problems, previous research has employed a range of feature selection strategies and machine-learning techniques. Using a dataset assembled from UCI, IEEE Data Port, and Kaggle, Uddin et al. [7] trained eleven ML algorithms, such as SVM, RF, and DT to identify cardiac conditions. With the DT, they achieved an astounding 99.16% accuracy rate. Nevertheless, overfitting, such high accuracy may indicate a situation where the model performs well on training data but poorly on unknown data. By merging Huang's method with k-mode clustering and utilizing methods like XGBoost, DT, RF, and Multilayer Perceptron, Javed et al. [8] aimed to increase classification accuracy. They tweaked the hyperparameters using GridSearchCV. With AUCs above 0.9, MLP was the best model, especially after cross-validation. The issue with this approach is that little variations in accuracy between cross-validation and non-cross-validation suggest that the model's performance could be erratic, thereby jeopardizing its reliability.

Chandrasekhar [9] developed a diagnostic method that optimized using GridSearchCV and five-fold cross-validation. After testing six classifiers, they found that AdaBoost and Logistic Regression had respective accuracy rates of 89.67% and 90.16%. The accuracy of the soft voting ensemble classifier was as high as 95%. Bhatt et al. [10] divided blood pressure data into intervals and translated ages to year bins prior to using k-mode clustering to a dataset on heart disease. At 87.23%, MLP had the highest accuracy. In this case, accuracy changes brought about by the study's preprocessing steps and clustering approach suggest that the model might not generalize well across a variety of datasets.

Ledig et al. [11] created a decision support tool that uses SMOTE for data balance, Hyperband for hyperparameter optimization, and very randomized trees for classification. The accuracy of their model was 98.52% on the Statlog dataset and 99.2% on the Cleveland dataset. This study's exceptionally high accuracy raises concerns about possible overfitting, which is a drawback. Additionally, even though tree-based models did well, more research was needed to determine how applicable they were to various datasets and real-world situations. These findings highlight the potential of ML in CVD prediction, but they also stress how important it is to address overfitting and validate models over a range of datasets to guarantee solid and trustworthy outcomes.

R. Subramanian and colleagues [12] presented utilizing neural networks to diagnose and forecast blood pressure, heart disease, and other traits. The output perceptron, which had about 120 hidden layers, was able to produce an output by incorporating the disease-related information into a deep neural network. If we use the model for the test dataset, this is the most basic and relevant way to ensure an accurate result of heart disease. It has been suggested that the supervised network be used to diagnose cardiac conditions [13]. The model that was employed and trained using the previously learnt data and forecasted the outcome when a doctor tested it using unknown data, determining the model's accuracy. A larger and more varied dataset provides a wider variety of examples for training models. Additionally, the majority of earlier studies used this experiment with a single heart disease dataset. We used a novel mixed heart disease dataset in this study, which consists of three different datasets. We proposed soft voting techniques based on enhancing a number of ML classifiers, including DT, RF, and LGBM. The outcome is improved by combining the RF and LGBM classifiers in an ensemble ML approach.

III. Metarial And Methods

The main goal of this method is to use medical information (age, blood pressure, cholesterol, etc.) to predict a person's risk of experiencing cardiac problems. This allows for a prompt and precise diagnosis of the issue. Five ML algorithms—SVM, RF, DT, KNN, and LGBM—were applied in this study. The proposed study assessed many categorization techniques and examined their effectiveness in predicting heart illness. The examination provides a precise evaluation of the person's cardiac condition. The data is entered by the doctor from the patient's medical file. An information-based strategy is used to estimate the likelihood of acquiring heart disease. The complete process is shown in Fig. 1.

A. Dataset

This study projected cardiac disease using three datasets. We have collected these three datasets. Two datasets from Kaggle [14,15] and one from IEEE Data Port [16] are included. The gathered dataset contains one targeted variable, thirteen attribute values, and fourteen features, despite the study's 5450 sample size. TABLE I shows the total number of entries from three different data sets along with the number of attributes and characteristics.

TABLE I. Datasets For Heart Disease

SL. NO.	Dataset Name	Attributes	Total Entry
1	Heart Disease Predication Dataset, Kaggle [15]	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, sl_slope, target	1190
2	Heart Disease Predication, Kaggle [14]	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, slope, target.	3235
3	Heart Disease Dataset (Comprehensive),IEEE Data port[16]	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target	1025

To identify cardiac illness, each of the three datasets employed distinct characteristics. It includes the patient's age, gender, and kind of chest pain (0 asymptomatic, 1 rare angina, 2 non-anginal discomfort, and 3 typical angina). Additionally, it logs resting ECG readings (0 for normal, 1 for STT wave abnormalities, and 2 for left ventricular hypertrophy), standing blood pressure levels (in mmHg), serum cholesterol levels (in mg/dl), and fasting blood sugar levels above 120 mg/dl (1 for true, 0 for false). Additional features include activity-induced angina (0 for no, 1 for yes), exercise-induced ST depression, the highest heart rate

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attained, and the slope of the peak workout ST segment (0 The presence of heart illness (0 no, 1 yes), thalassemia (3 normal, 6 fixed defect, 7 reversible defect), number of major vessels (0-3), inclining, 1 level, and 2 sloping.

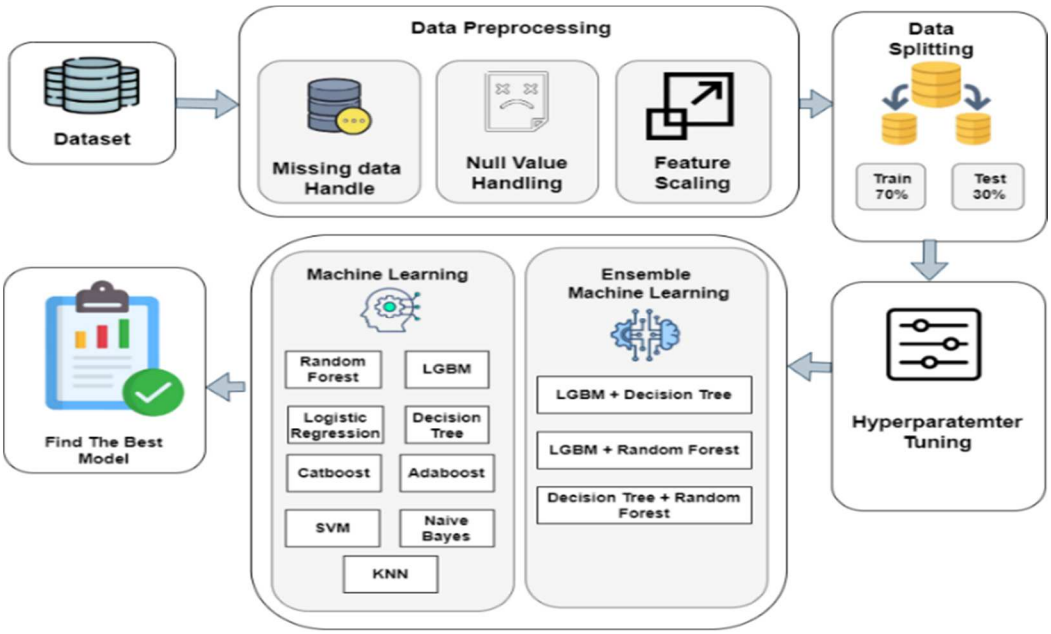


Fig-1 Proposed model for heart disease prediction

B. Data-Preprocessing

Before supplying the dataset to ML models, the data preparation stage of our study is essential for guaranteeing the dataset's quality and integrity. Here, we go over the precise procedures we used to prepare the data for heart disease prediction. We looked at a range of hyperparameters for every model using Grid Search with Cross-Validation (CV). tree depth is 20, and the smallest sample size per leaf is 5. The highest cross-validation score for RF is 0.9354 and for LGBM is 0.9482. Similarly, for LGBM the learning rate is 0.1, there are 200 estimator trees with 31 leaves, and the boosting type is GBDT.

Handling Null Value- Missing values must be handled carefully during data preparation. Our dataset had several missing data points, which might have led to inaccurate estimates. Imputation, which substitutes the median values of the respective characteristics for missing values, allowed us to overcome this problem. This approach helps to prevent biases and maintain data consistency.

Missing Data Handling- Data entry errors or missing patient records may be the cause of missing data in medical databases. We employed the median imputation technique to get around this. Missing data in numerical variables including age, blood pressure, and cholesterol levels were filled in using median imputation. The median is a more dependable option for handling skewed distributions, which are typical in medical data, because it is less vulnerable to outliers than the mean.

Data Splitting- A crucial component of our strategy was dividing the data into training and testing sets. The remaining 70% of the data was used for training, while 30% was kept for assessment.

Feature Scaling: Because the dataset included features in a range of units (e.g., age in years, cholesterol in mg/dl), feature scaling was required to guarantee that each characteristic contributed equally to the model's learning process. Standardization was selected over normalization because it converts the data to a mean of zero and a standard deviation of one, which is very useful for feature-sensitive models. These preparatory methods were applied to enhance model performance and preserve data quality. Because outliers are common in medical datasets, median imputation was selected for its handling capabilities. Standardization of feature scaling enhances model performance and convergence speed.

C. ML Algorithms

RF:

Regression analysis and classification are done using random-forest methods. For forecasting purposes, data is shown as a tree. On large datasets, RF may still provide reliable results even if some record entries are missing. Several data sets may benefit from the examination and study of the DT. There are two steps to a RF: first, it creates the RF, and then it projects the classification that was determined in the first stage. Equation (1) determines the likelihood that an object will be classified as belonging to a particular class or feature using pi.

$$Gini = 1 - \sum_{i-1}^{n}(pi)^2 \quad (1)$$

Finding a balance between accuracy, interpretability, computational efficiency, and resilience against overfitting was the main goal of the selection criterion.

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D. Environment Setup

Data was gathered, divided into assessment and training sets, and then injected at random. Some input-prepping techniques were only performed during the validation phase prior to being applied to the test sets in order to prevent input losses and overfitting. Using Google Co-Lab and Python modules like Scikit-Learn, NumPy, Pandas, Seaborn, and Matplotlib, the entire procedure took place remotely on a personal computer. The essential elements and system requirements for the suggested model are displayed in TABLE II.

TABLE II. SYSTEM SPECIFICATION OF THE PROPOSED MODEL

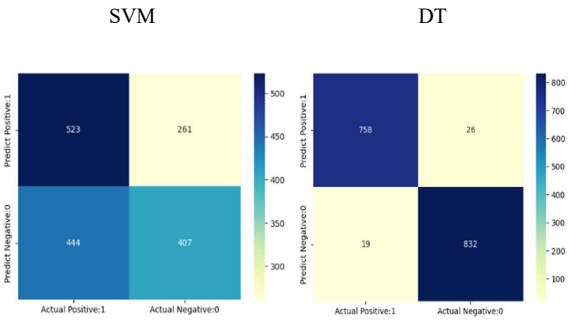
System requirements for ML algorithm development	
RAM	12.68 GB
CPU	2- core xeon 2.2 GHz
Chache	56320KB
GPU	Tesla K80 GPU
GPU Memory	12 GB

IV. RESULT ANALYSIS AND DISCUSSION

The experimental results, presented in Table IV, provide a comprehensive assessment of the performance of several ML algorithms in terms of accuracy, precision, recall, and F1 score. With the greatest accuracy of 98.89%, precision, recall, and F1 scores of 99%, 98%, and 99%, respectively, RF outperformed the other classifiers, proving its resilience in the face of the classification job. With an accuracy of 98.16%, precision of 99%, recall of 96%, and F1 score of 97%, DT demonstrated good performance as well, demonstrating its effectiveness and dependability as a model. However, with accuracies of 88.99% and 86.60%, respectively, models like KNN and SVM performed very poorly, suggesting that these classifiers might not be as well suited for this dataset. Both CatBoost and LGBM demonstrated strong performance; CatBoost achieved a little lower accuracy of 96.88% with comparable precision and recall, while LGBM achieved an accuracy of 97.16% with precision, recall, and F1 scores of 97%.

TABLE III. FACTORS COMPRISING THE CONFUSION MATRIX

Actual Values			
	Negative(0)	Positive (1)	
Negative(0)	TN	FN	Predicted
Positive (1)	FN	TN	



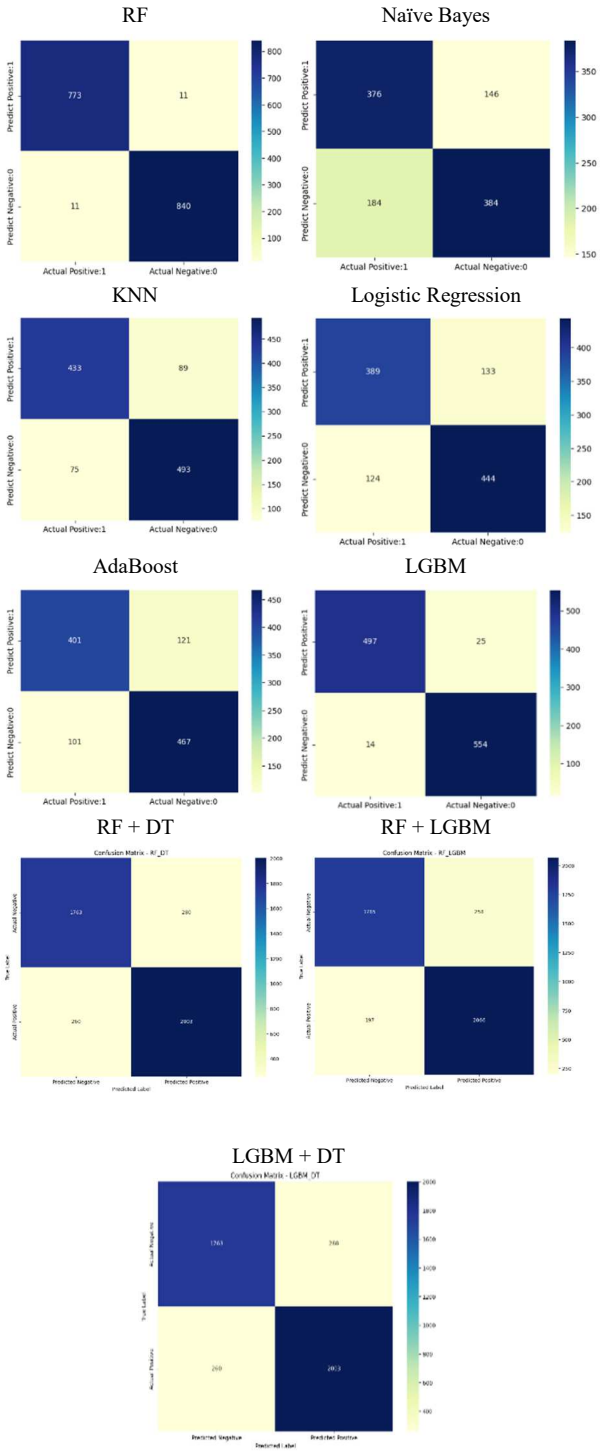


Fig- 2. Confusion matrices for several ML classifiers.

Ensemble models such as LGBM+DT and LGBM+RF significantly improved the classification performance, achieving 97.88% accuracy and a 99% F1 score for LGBM+DT and 98.44% accuracy and a 97% F1 score for LGBM+RF, respectively. The DT and RF (DT+RF) combination demonstrated that while the ensemble approach enhanced performance, it did not always surpass the individual models on all metrics, with an accuracy of 97.88% and an F1 score of 95%.

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TABLE IV. MODEL ANALYSIS OF ML ALGORITHM AND THE RESULTS

Model	Accuracy	Precision	Recall	F1 Score
DT	98.16%	99%	96%	97%
RF	98.89%	99%	98%	99%
KNN	88.99%	88%	90%	89%
SVM	86.60%	85%	89%	87%
LGBM	97.16%	97%	97%	97%
CatBoost	96.88%	98%	96%	97%
AdaBoost	78.44%	95%	96%	97%
NB	69.63%	93%	98%	98%
LR	77.88%	99%	95%	95%
LGBM+DT	97.88%	99%	96%	99%
LGBM+RF	98.44%	99%	96%	97%
DT+RF	97.88%	98%	97%	95%

The results are further supported by the examination of Receiver Operating Characteristic (ROC) curves, which are displayed in Figures 3 and 4. Given their bigger areas under the curve (AUC) and greatest true positive rates, the results show that RF and DT classifiers are more effective at distinguishing between the positive and negative classes. Furthermore, the confusion matrices shown in Table III and Fig. 2 demonstrate that the RF and DT models demonstrated balanced classification outcomes with few false positives and false negatives, further enhancing their efficacy, in addition to achieving high overall accuracy.

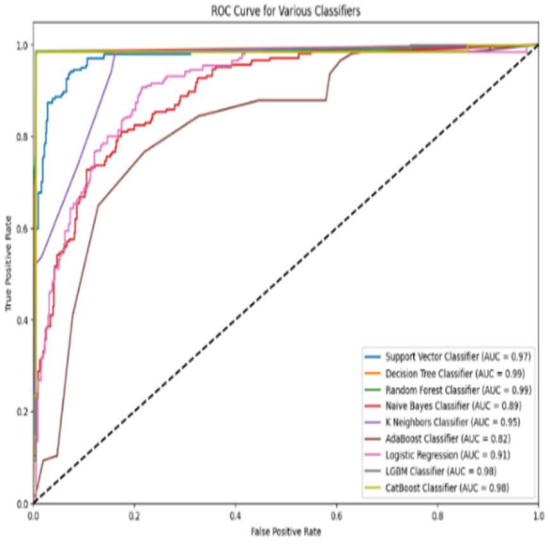


Fig -3 ROC curve for individual classifiers

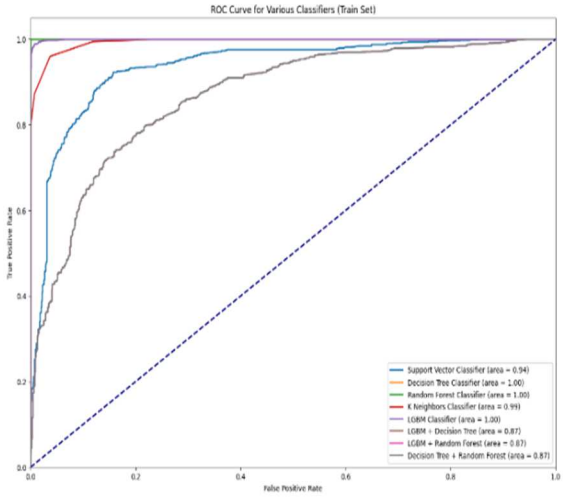


Fig- 4 ROC curve of several individual and ensemble classifiers

These findings highlight the benefit of combining classifiers to improve model performance in challenging classification tasks, showing that while individual classifiers such as RF and DT offer high classification accuracy, ensemble models such as LGBM+DT and LGBM+RF offer additional improvements in recall and F1 score.

TABLE V. MODEL ANALYSIS OF ML ALGORITHM AND THE RESULTS

Author	Dataset	Method	Accuracy
Chandrasekhar et al. [9]	Cleveland	Soft voting ensemble method	93.44%
Bharti et al. [17]	UCI	Deep Learning	94.2%
Maini et al.[18]	Cleveland Heart Disease	ANN	90.74%
Proposed Method	Kaggle+Kaggle+IEEE	RF	98.89%

The efficiency of the proposed RF-based model is contrasted with existing methods in related research, as shown in Table V. With an accuracy of 93.44%, Chandrasekhar et al. [9] used a soft voting ensemble approach on the Cleveland dataset. On the UCI dataset, Bharti et al. [17] applied deep learning algorithms, which marginally increased the accuracy to 94.2%. The Cleveland Heart Disease dataset was subjected to an Artificial Neural Network (ANN) by Maini et al. [18], who achieved an accuracy of 90.74%. On the other hand, the suggested approach, which made use of the RF algorithm and was evaluated on a variety of datasets that combined the IEEE and Kaggle datasets, produced a noticeably better accuracy of 98.89%. This significant improvement highlights how well the suggested method performs classification tasks, even when dealing with datasets that include intricate patterns and imbalances. RF's remarkable success was probably influenced by its capacity to handle high-dimensional data, minimize overfitting, and operate stably with little parameter adjustment, in contrast to more conventional ensemble techniques and neural network approaches. In addition, compared to earlier research that depended on a single, smaller dataset, the suggested method's use of a more representative and diversified dataset guaranteed more generalizability and improved prediction accuracy. The outcomes unequivocally show that the suggested model not only performs better than earlier approaches but also establishes a new standard for classification accuracy in this field, suggesting its potential for real-world uses and more developments in machine learning-based forecasting.

V. CONCLUSIONS

Since heart disease is still one of the leading causes of mortality worldwide, accurate and effective prediction models are essential. With the RF classifier achieving 98.89% accuracy, 99% precision, 98% recall, and 99% F1 score, this study demonstrated the exceptional efficacy of ML techniques for the prediction of heart disease. We greatly improved the models' dependability and generalizability by implementing sophisticated preprocessing techniques, hyperparameter tweaking, and ensemble approaches. The suggested system's resilience was further enhanced by combining three different datasets, which decreased

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overfitting and enhanced classification metrics generally. Future research will concentrate on enhancing the models' interpretability through the use of explainable AI (XAI) methods like LIME and SHAP. These methods will help physicians better comprehend and trust the forecasts by offering insights into feature contributions. Additionally, adding real-time clinical data to the dataset and investigating hybrid ensemble approaches may improve the suggested system's scalability and flexibility, making it more appropriate for practical clinical applications. This study lays the groundwork for open and effective healthcare solutions while also representing a major breakthrough in machine learning-driven heart disease prediction.

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